

Experiment 1

Basic Image Processing and Image Enhancement using Point Processing

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

img=cv2.imread('sample.jpg')

img_rgb=cv2.cvtColor(img,cv2.COLOR_BGR2RGB)

cv2.imwrite('output.png',img)

gray=cv2.cvtColor(img,cv2.COLOR_BGR2GRAY)

blur=cv2.GaussianBlur(img,(5,5),0)

equalized=cv2.equalizeHist(gray)

kernel=np.array([[[-1,-1,-1],
                  [-1,5,-1],
                  [-1,-1,-1]]])

sharp=cv2.filter2D(img,-1,kernel)

titles=['Original','Gray','Blur',
        'Equalized','Sharpened']

images=[img_rgb,gray,blur,equalized,sharp]

plt.figure(figsize=(15,5))

for i in range(5):

    plt.subplot(1,5,i+1)

    if len(images[i].shape)==2:
        plt.imshow(images[i],cmap='gray')

    else:
        plt.imshow(images[i])

    plt.title(titles[i])

    plt.axis('off')

plt.show()
```

Viva Questions

Q1. cv2.cvtColor() use?

Image color conversion ke liye.

Q2. Histogram equalization use?

Contrast enhancement.

Q3. Gaussian blur purpose?

Noise reduction.

Q4. filter2D() kya karta hai?

Custom kernel apply karta hai.

Experiment 2

Image Enhancement using Spatial Filtering

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

img=cv2.imread('sample.jpg',0)

kernel=np.ones((5,5),np.float32)/25

convolution=cv2.filter2D(img,-1,kernel)

average=cv2.blur(img,(5,5))

gaussian=cv2.GaussianBlur(img,(5,5),0)

median=cv2.medianBlur(img,5)

sobelx=cv2.Sobel(img,cv2.CV_64F,1,0,ksize=3)

sobely=cv2.Sobel(img,cv2.CV_64F,0,1,ksize=3)
```

```

laplacian=cv2.Laplacian(img,cv2.CV_64F)

log=cv2.GaussianBlur(img,(3,3),0)

log=cv2.Laplacian(log,cv2.CV_64F)

titles=['Original','Convolution','Average',
        'Gaussian','Median','SobelX',
        'SobelY','Laplacian','LoG']

images=[img,convolution,average,
        gaussian,median,sobelx,
        sobely,laplacian,log]

plt.figure(figsize=(18,8))

for i in range(9):

    plt.subplot(3,3,i+1)

    plt.imshow(images[i],cmap='gray')

    plt.title(titles[i])

    plt.axis('off')

plt.show()

```

Viva Questions

Q1. Median filter use?

Salt and pepper noise removal.

Q2. Sobel operator use?

Edge detection.

Q3. Laplacian detects?

Second-order edges.

Q4. LoG full form?

Laplacian of Gaussian.

Experiment 3

Image Segmentation using Thresholding and K-Means Clustering

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

img=cv2.imread('sample.jpg')

img_rgb=cv2.cvtColor(img,cv2.COLOR_BGR2RGB)

gray=cv2.cvtColor(img,cv2.COLOR_BGR2GRAY)

ret,thresh=cv2.threshold(gray,127,255,
                        cv2.THRESH_BINARY)

pixel_values=img.reshape((-1,3))

pixel_values=np.float32(pixel_values)

criteria=(cv2.TERM_CRITERIA_EPS+
          cv2.TERM_CRITERIA_MAX_ITER,
          100,0.2)

k=3

retval,label,center=cv2.kmeans(
    pixel_values,
    k,
    None,
    criteria,
    10,
    cv2.KMEANS_RANDOM_CENTERS)

center=np.uint8(center)

segmented=center[label.flatten()]

segmented_image=segmented.reshape(img.shape)

titles=['Original','Grayscale',
        'Threshold','K-Means Segmentation']
```

```
images=[img_rgb,gray,thresh,segmented_image]
```

```
plt.figure(figsize=(12,5))
```

```
for i in range(4):
```

```
    plt.subplot(1,4,i+1)
```

```
    if len(images[i].shape)==2:
```

```
        plt.imshow(images[i],cmap='gray')
```

```
    else:
```

```
        plt.imshow(images[i])
```

```
    plt.title(titles[i])
```

```
    plt.axis('off')
```

```
plt.show()
```

Viva Questions

Q1. Thresholding kya hai?

Binary segmentation technique.

Q2. K-means kya karta hai?

Pixels ko clusters me divide karta hai.

Q3. K value kya hai?

Number of clusters.

Q4. Segmentation purpose?

Object separation.

Experiment 4

Image Interest Point Detection using Harris Corner Detector

```

import cv2
import numpy as np
import matplotlib.pyplot as plt
from scipy.ndimage import maximum_filter

gray=cv2.imread('sample.jpg',
                cv2.IMREAD_GRAYSCALE)

gray_float=np.float32(gray)

Ix=cv2.Sobel(gray_float,cv2.CV_64F,
             1,0,ksize=3)

Iy=cv2.Sobel(gray_float,cv2.CV_64F,
             0,1,ksize=3)

Ixx=Ix*Ix
Iyy=Iy*Iy
Ixy=Ix*Iy

Sxx=cv2.GaussianBlur(Ixx,(5,5),1.5)

Syy=cv2.GaussianBlur(Iyy,(5,5),1.5)

Sxy=cv2.GaussianBlur(Ixy,(5,5),1.5)

det_M=(Sxx*Syy)-(Sxy**2)

trace_M=Sxx+Syy

R_cv=det_M-0.04*(trace_M**2)

corner_mask=R_cv>0.01*R_cv.max()

local_max=maximum_filter(R_cv,size=10)

corner_nms=(R_cv==local_max)&corner_mask

corner_y,corner_x=np.where(corner_nms)

img_bgr=cv2.cvtColor(gray,cv2.COLOR_GRAY2BGR)

for (x,y) in zip(corner_x,corner_y):

    cv2.circle(img_bgr,(x,y),
               5,(0,0,255),-1)

```

```

img_rgb=cv2.cvtColor(img_bgr,
                    cv2.COLOR_BGR2RGB)

titles=['Original','Gradient lx',
        'Gradient ly',
        'Harris Response',
        'Detected Corners']

images=[gray,lx,ly,R_cv,img_rgb]

plt.figure(figsize=(18,5))

for i in range(5):

    plt.subplot(1,5,i+1)

    if i==4:
        plt.imshow(images[i])

    else:
        plt.imshow(images[i],cmap='gray')

    plt.title(titles[i])

    plt.axis('off')

plt.show()

print(f'Total corners detected : {len(corner_x)}')

```

Viva Questions

Q1. Harris detector use?

Corner detection.

Q2. Gradient lx and ly?

X and Y direction intensity change.

Q3. Corner kya hota hai?

Intensity both directions me change hoti hai.

Q4. Sobel operator role?

Gradient calculation.

Experiment 5

Object Recognition using HOG and Machine Learning

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

img=cv2.imread('sample.jpg')

img_rgb=cv2.cvtColor(img,
                    cv2.COLOR_BGR2RGB)

gray=cv2.cvtColor(img,
                  cv2.COLOR_BGR2GRAY)

orb=cv2.ORB_create()

kp,des=orb.detectAndCompute(gray,None)

img_features=cv2.drawKeypoints(
    img_rgb,
    kp,
    None,
    color=(0,255,0))

hog=cv2.HOGDescriptor()

h=hog.compute(gray)

titles=['Original Image',
        'Feature Detection']

images=[img_rgb,img_features]

plt.figure(figsize=(10,5))

for i in range(2):

    plt.subplot(1,2,i+1)

    plt.imshow(images[i])
```



```
plt.title(titles[i])

plt.axis('off')

plt.show()

print('Number of Keypoints :',len(kp))

print('HOG Feature Vector Shape :',h.shape)
```

Viva Questions

Q1. HOG full form?

Histogram of Oriented Gradients.

Q2. ORB use?

Feature detection and description.

Q3. Keypoints kya hote hain?

Important image points.

Q4. HOG use?

Object recognition and detection.